

→ Machine learning & its types (Supervised, Unsupervised, Reinforcement learning, Active learning)

→ Steps of Machine learning workflow with flow diagram

→ Challenges in ML & Ethical considerations.

→ Types of Supervised learning -

- Regression & its types & numerical problems.

- Classification → KNN

→ Numerical

- SVM & its different concepts like Hyperplane, Support vectors, kernels, Margin.

- Decision Tree

→ Unsupervised learning

Clustering & its types.

- k-Means

- DBSCAN

- Hierarchical

Agglomerative

Divisive

→ Dimensionality Reduction

- PCA

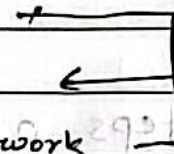
- LDA

Neural Networks Architectures:

- Feedforward

- Convolution

- Recurrent neural network



→ Their application

- Forward & backward ~~prop~~ propagation in neural network with suitable diagram.

- Perception & its types

→ Gradient descent

→ Model evaluation in ML & its consequences

• Evaluation metrics for classification & Regression

• Cross-validation & k-fold cross validation with diagram, & compare it with

• Train-Test Split